

ZARI.ai EDA Visual Analysis & Dataset Report

Complete Side-by-Side Comparison: Full 67-Class Dataset vs 3-Crop Target Dataset

1. Executive Summary: Dual Dataset Scope

This technical report presents the complete Exploratory Data Analysis (EDA) graphs and statistical distributions for both dataset paradigms in ZARI.ai:

- 1. Full Master Dataset (Phase 1): 67 disease classes across 14 crop species (87,000+ total leaf images).
- 2. Production 3-Crop Dataset (Phase 3): Focused target scope covering Tomato (13 classes), Potato (3 classes), and Pepper (6 classes) for 22 total canonical disease classes (24,500+ images).

2. Full Dataset EDA (67 Classes across 14 Crops)

Exploratory analysis of the full 67-class multi-crop dataset:

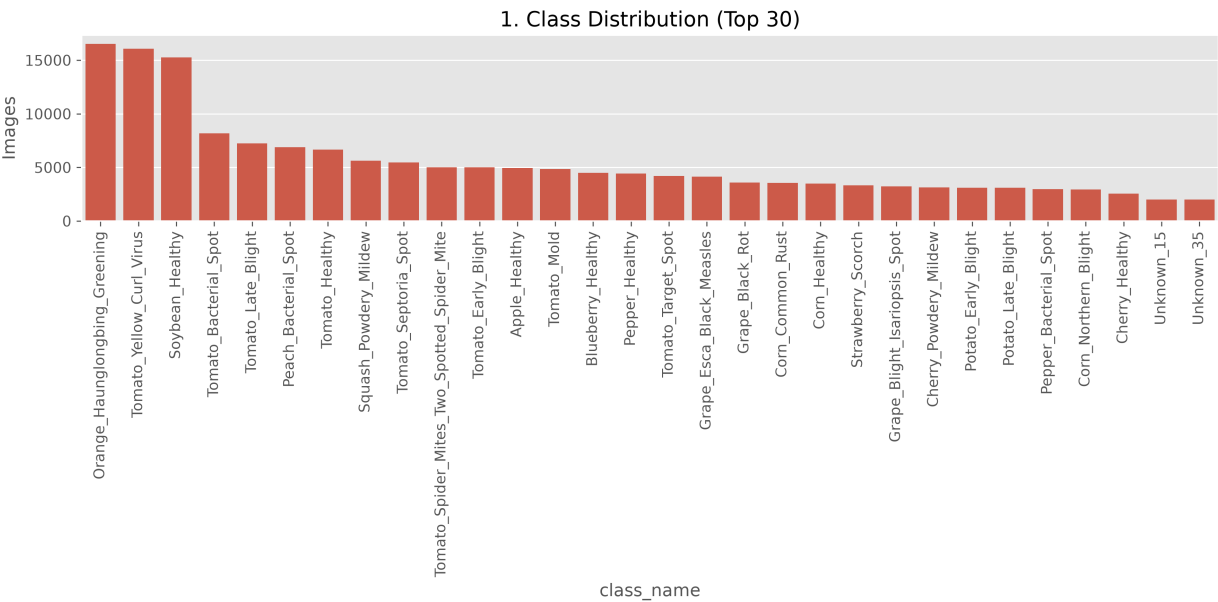


Figure: Full Dataset Master 67-Class Image Volume Distribution

2. Crop Distribution

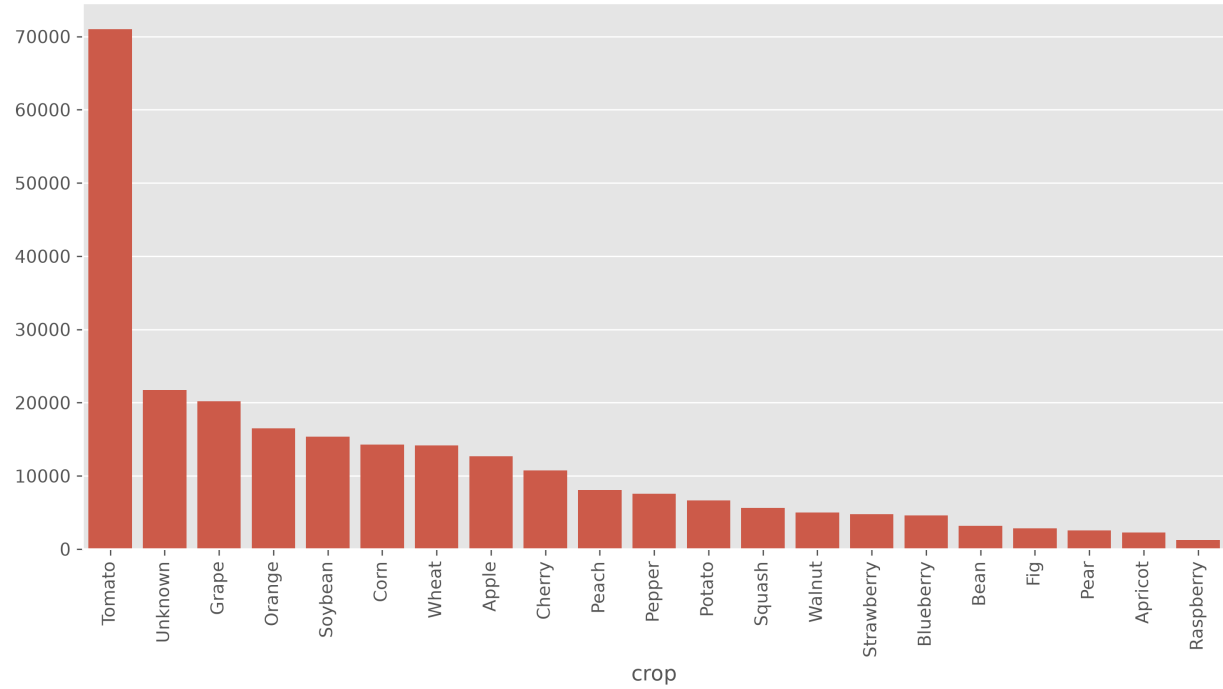


Figure: 14 Crop Species Breakdown Across Master Dataset

2. Full Dataset EDA (Continued)

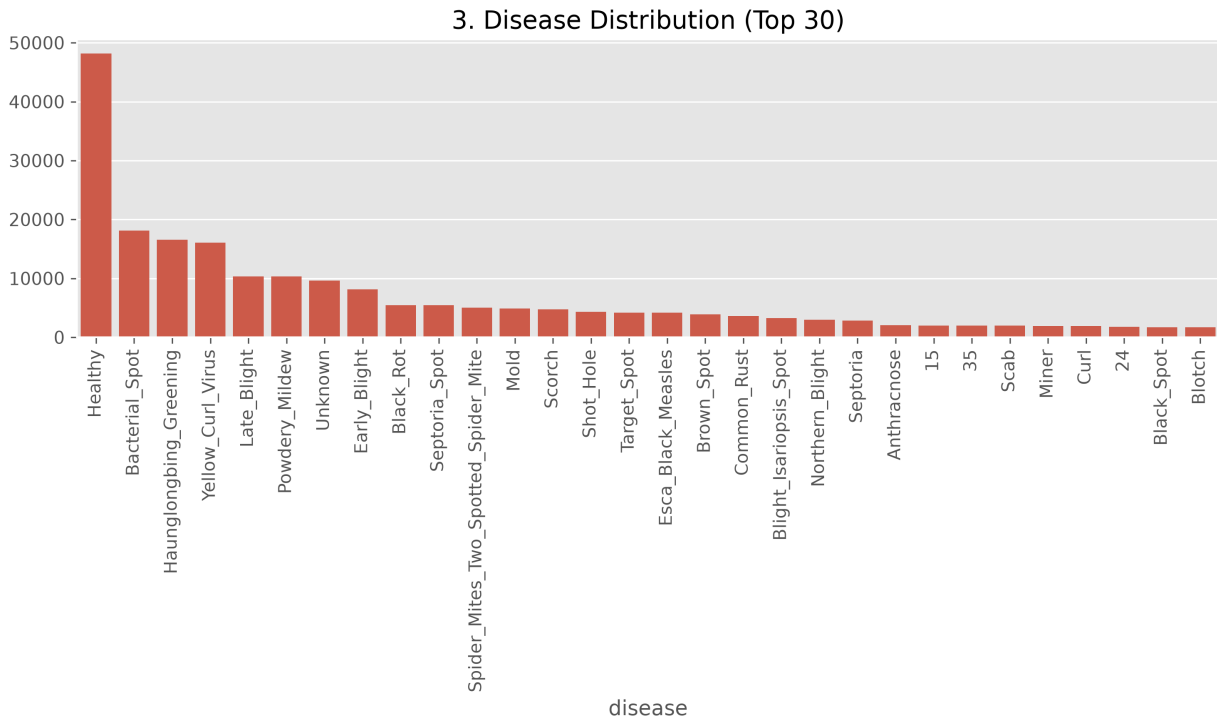


Figure: Healthy vs Diseased Leaf Sample Ratios

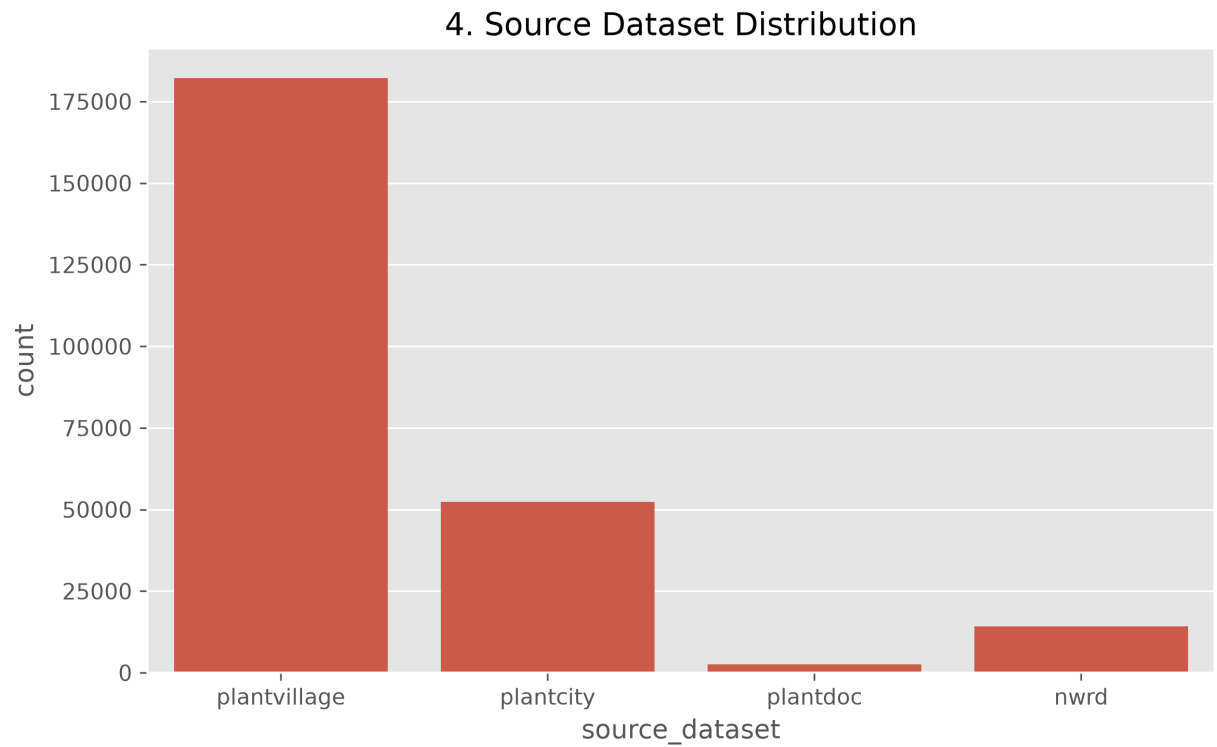


Figure: Multi-Source Data Provenance (PlantVillage, PlantDoc, Local Field Data)

5. Split Distribution

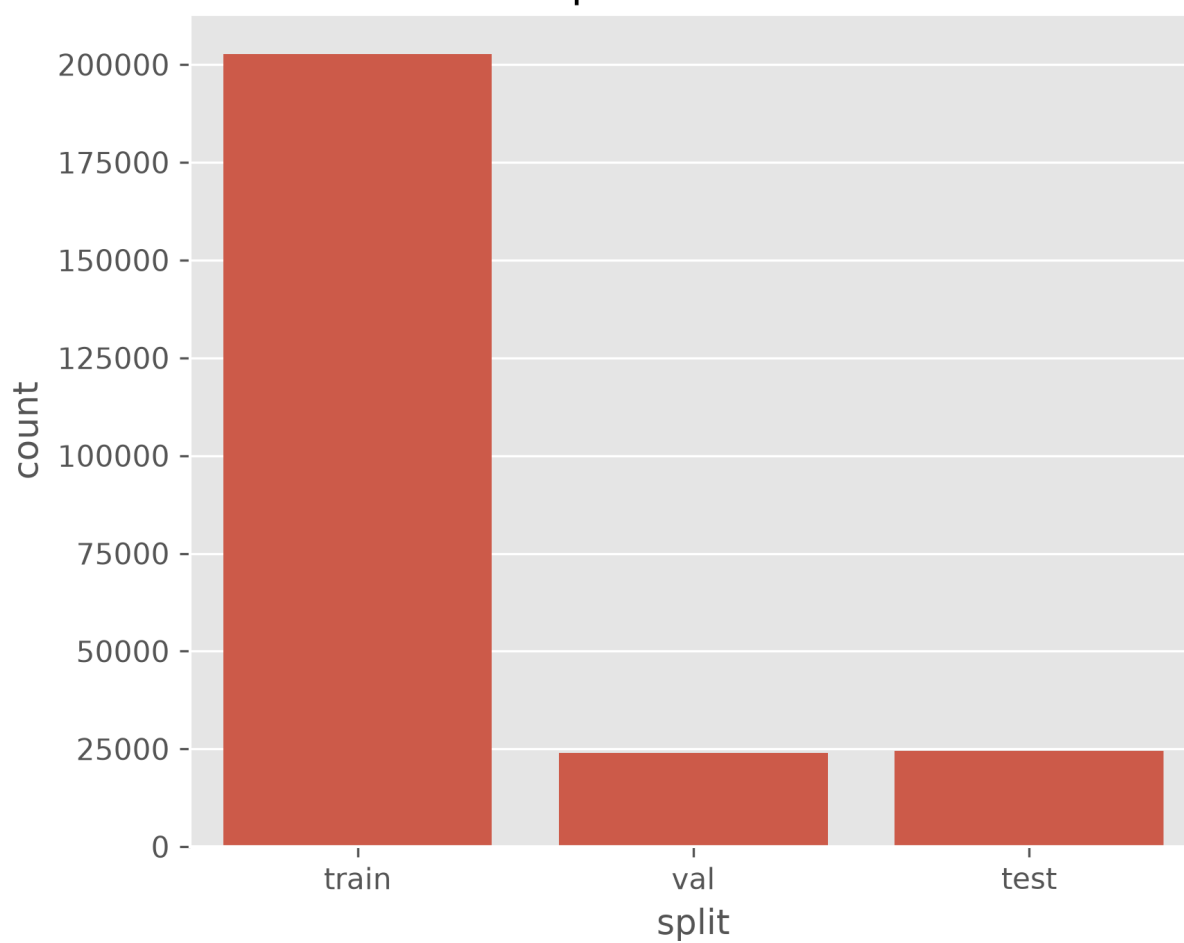


Figure: Stratified Train / Val / Test Split Distribution (80% / 10% / 10%)

6. Domain Distribution (Lab vs Field)

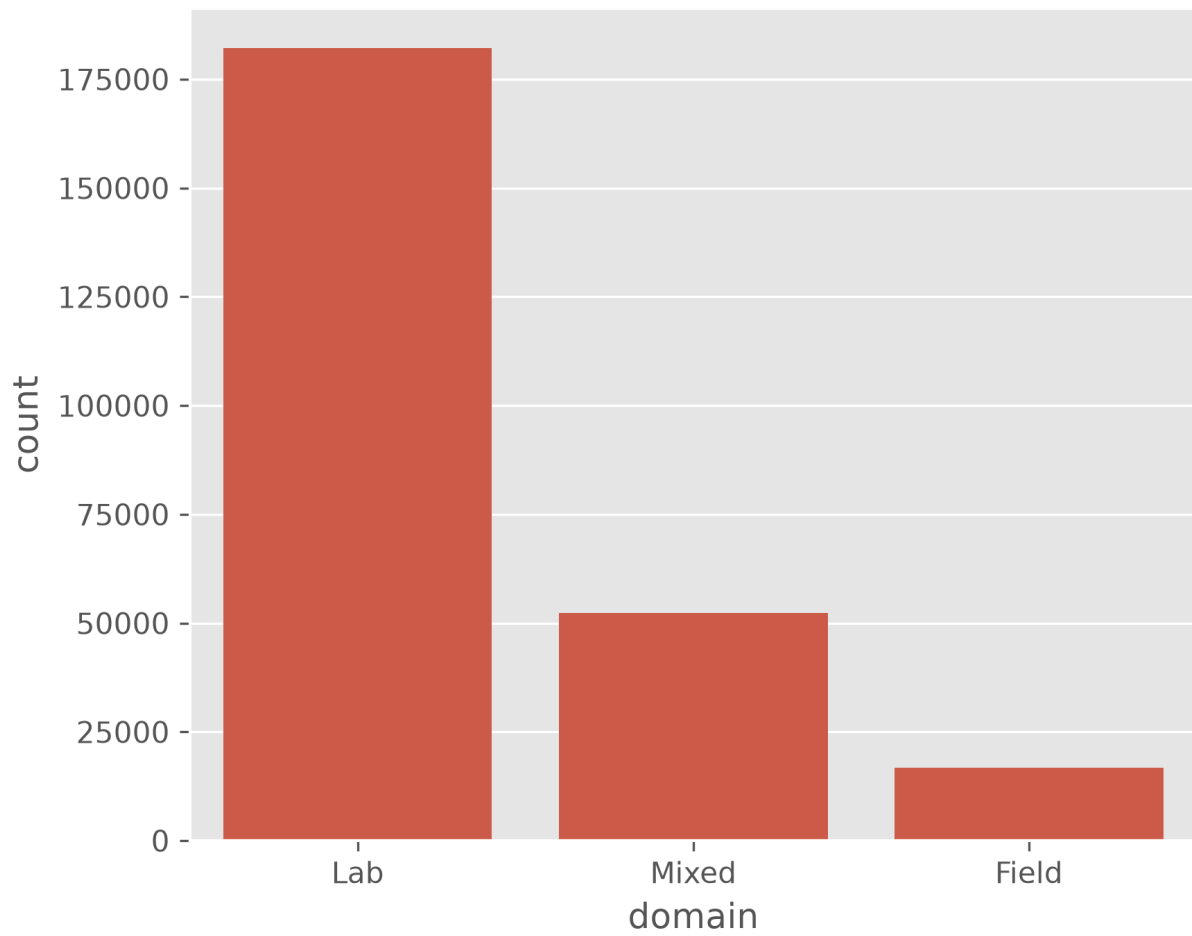


Figure: Lab-Controlled vs Field-Realistic Image Domain Split

7. Healthy vs Diseased

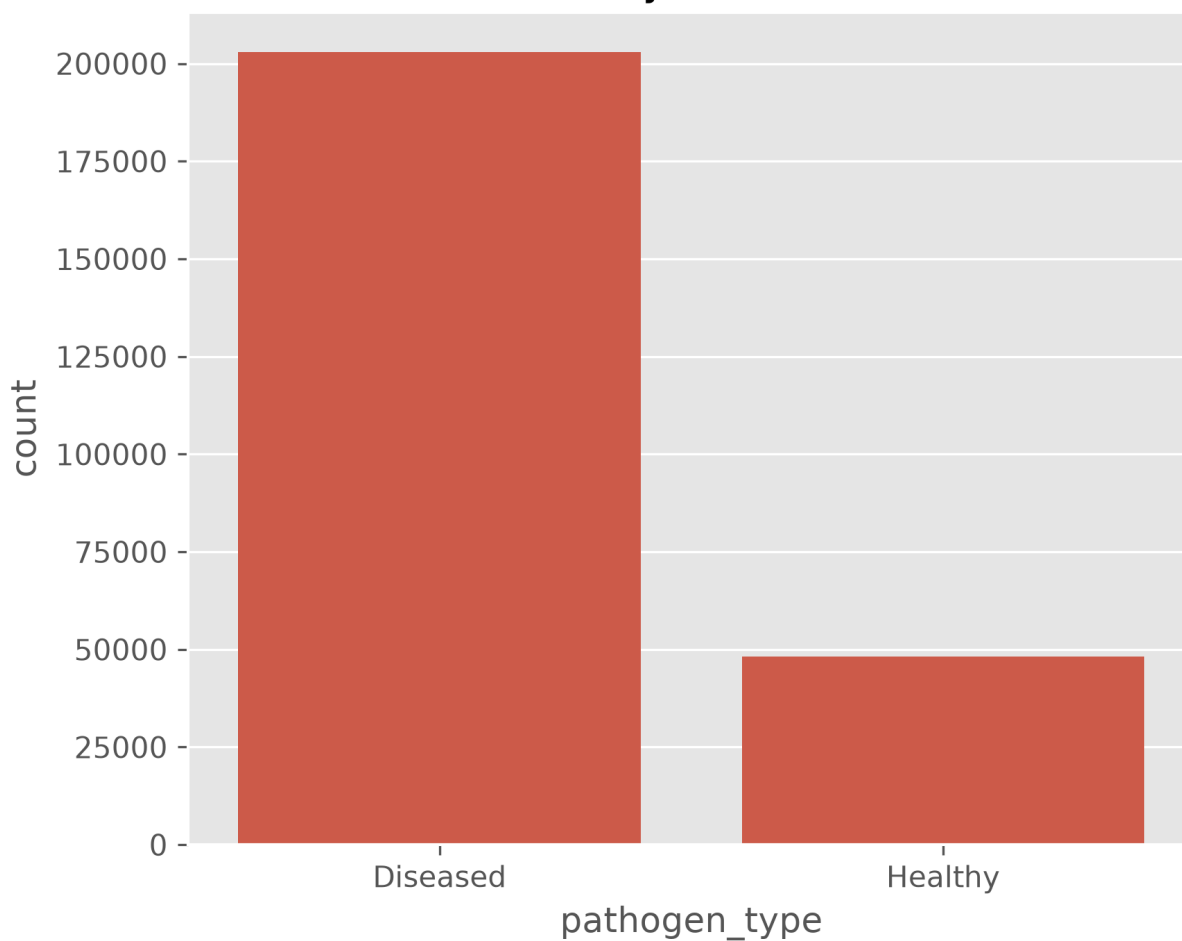


Figure: Crop-wise Healthy vs Diseased Sample Breakdown

8. Blur Score Distribution (Clipped)

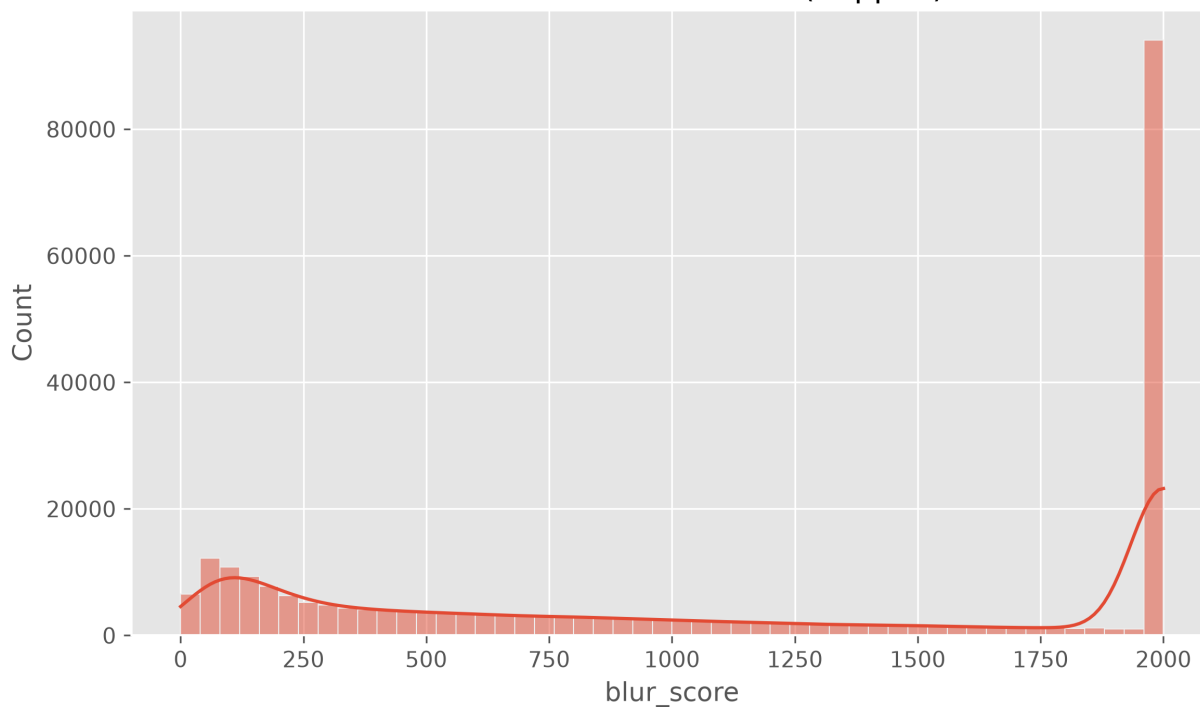


Figure: Laplacian Variance Blur Distribution (Quality Screening)

9. Brightness Distribution

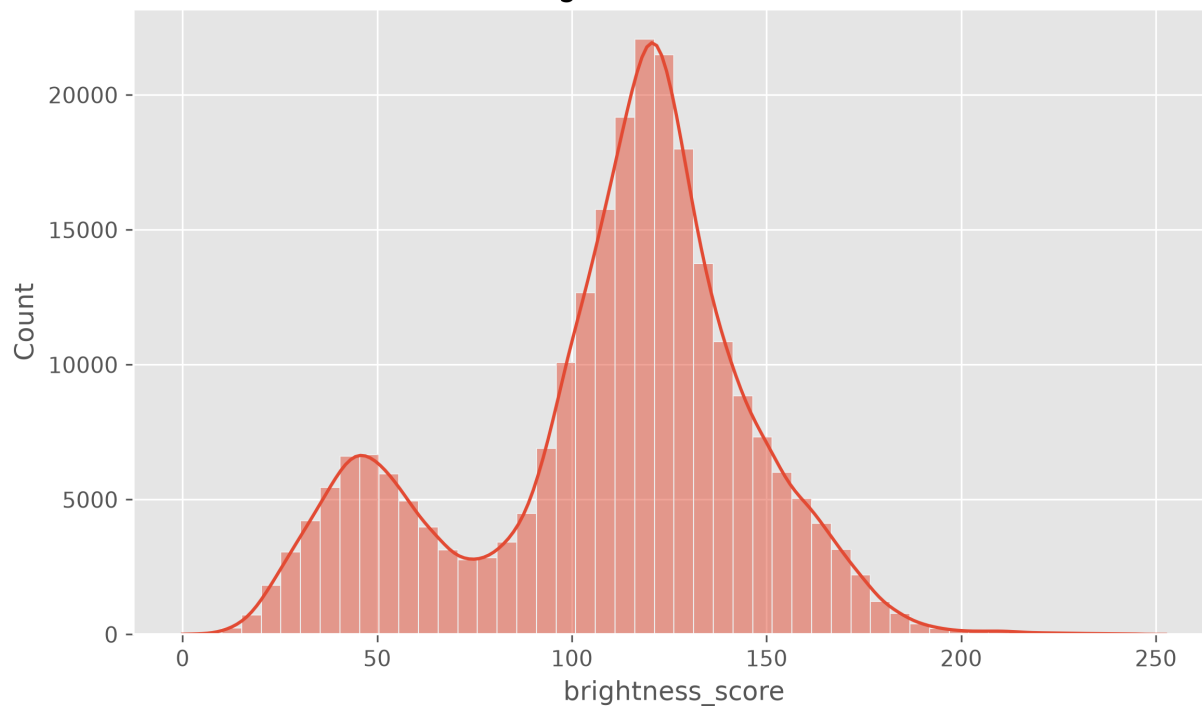


Figure: Mean Pixel Intensity / Brightness Distribution

10. Image Quality Distribution

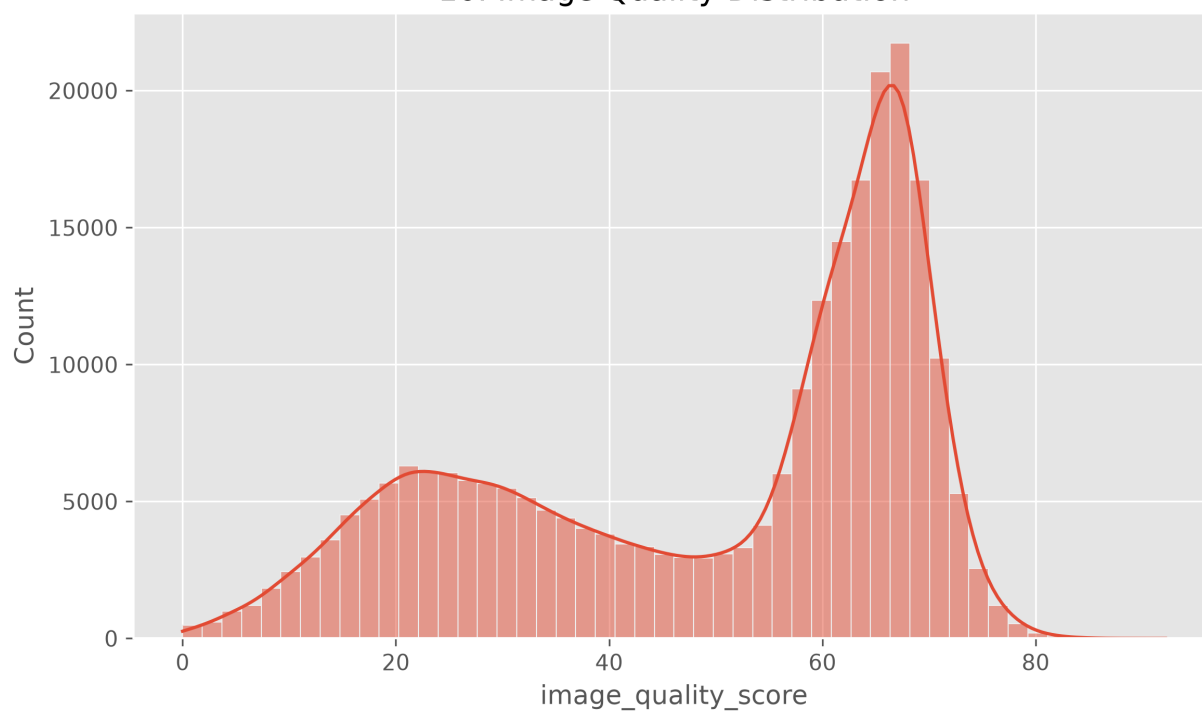


Figure: Signal-to-Noise Ratio (SNR) Quality Histogram

11. Image Difficulty Levels

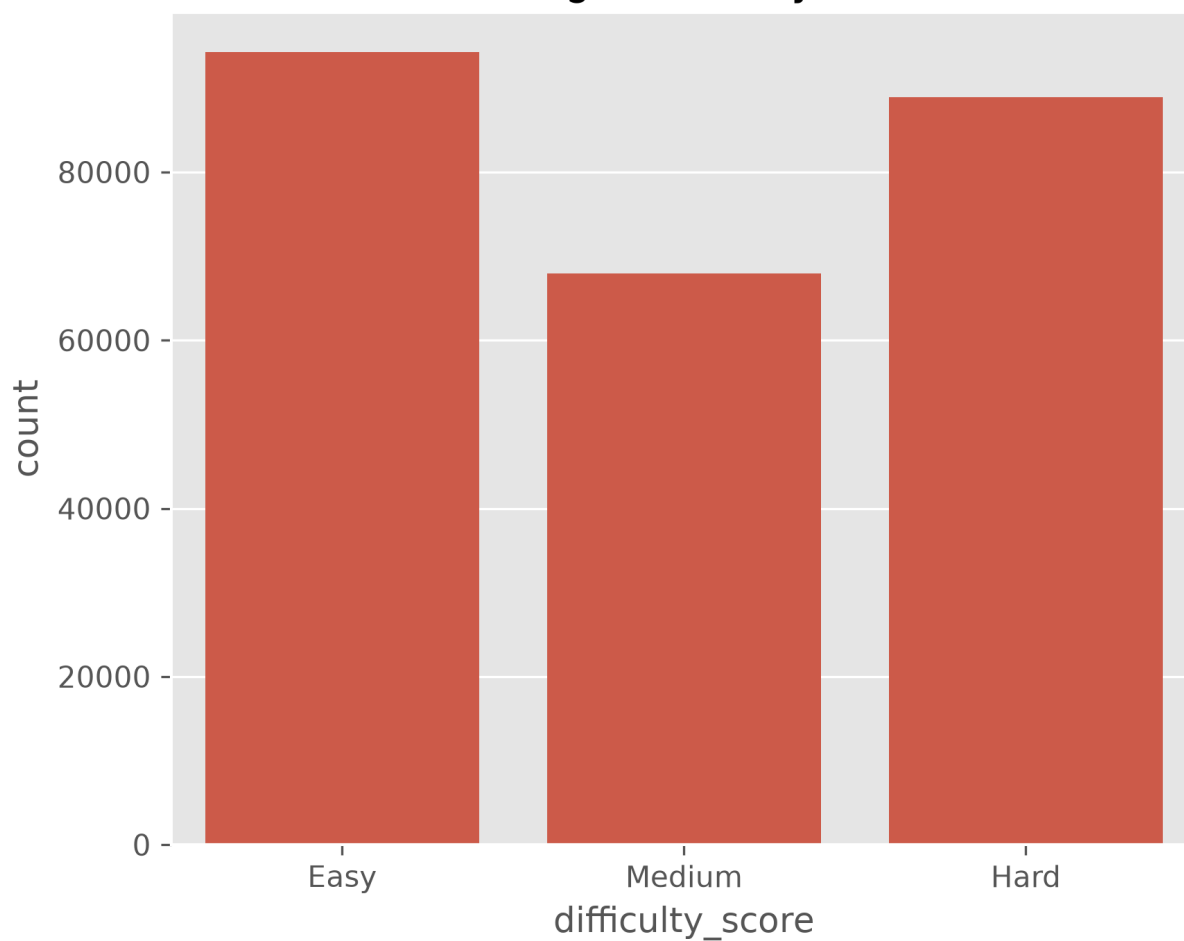


Figure: Sample Classification Difficulty Ratings

12. Severity Class Distribution

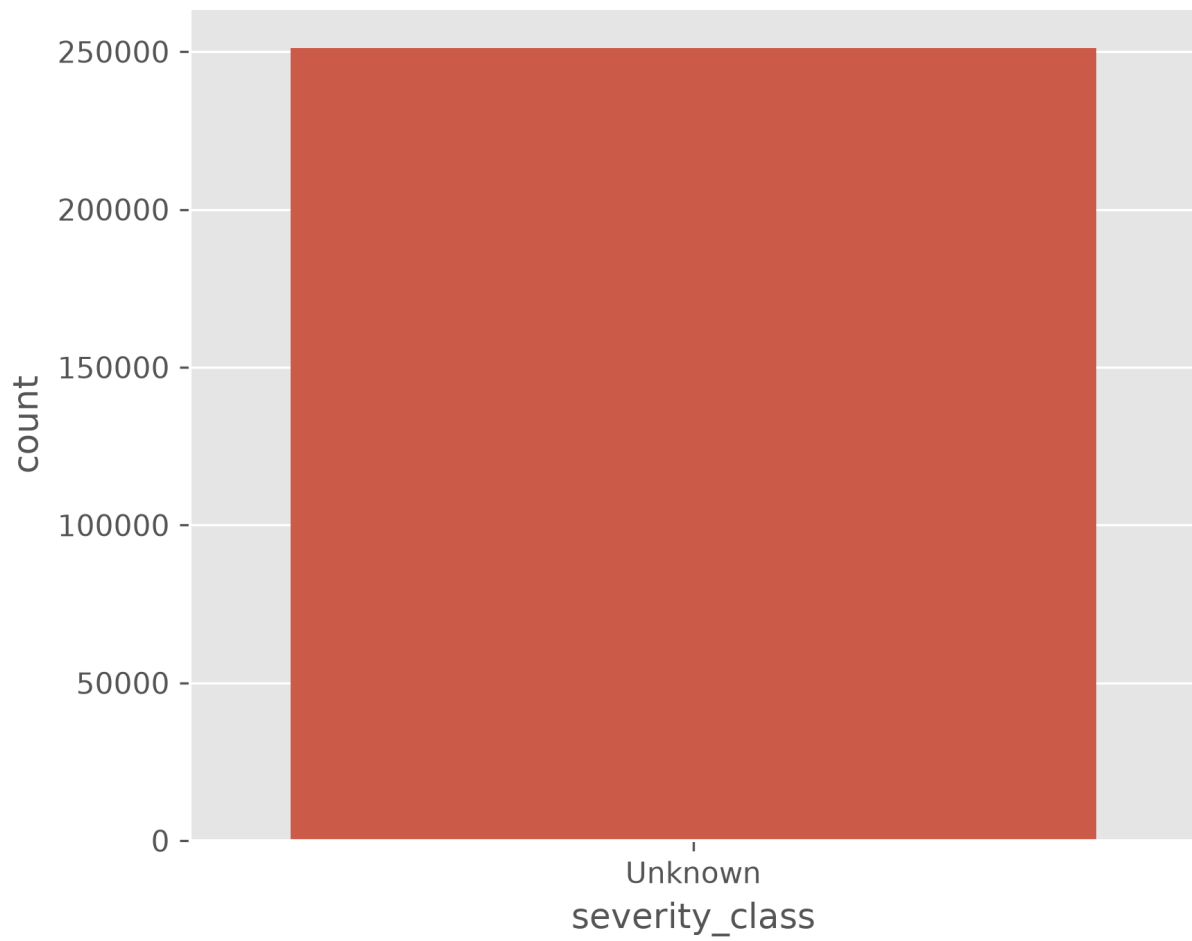


Figure: Infection Severity Breakdown (Early, Moderate, Severe)

13. Top 10 Classes Pie Chart

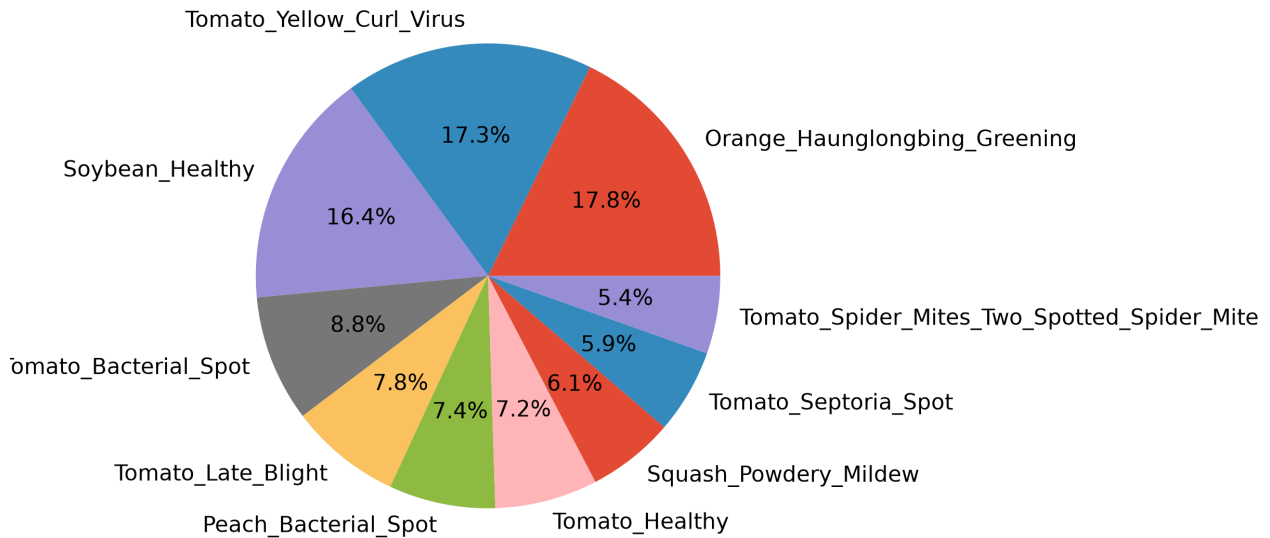


Figure: Top-10 Dominant Classes Volume Contribution

14. Bottom 30 Classes (Rare Classes)

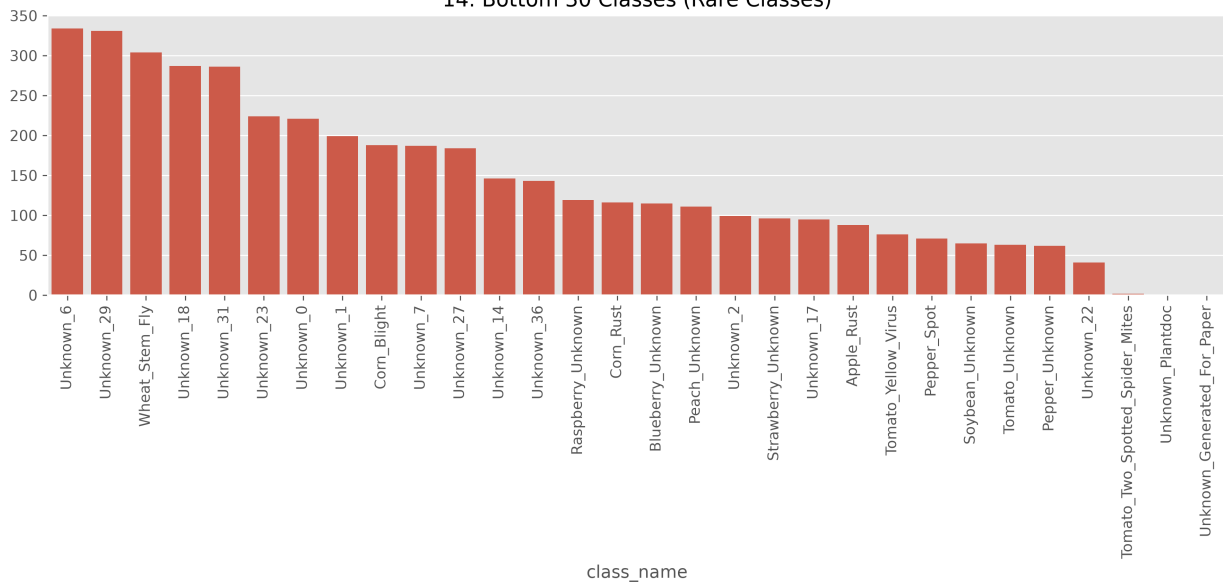


Figure: Rare & Long-Tail Class Imbalance Distribution

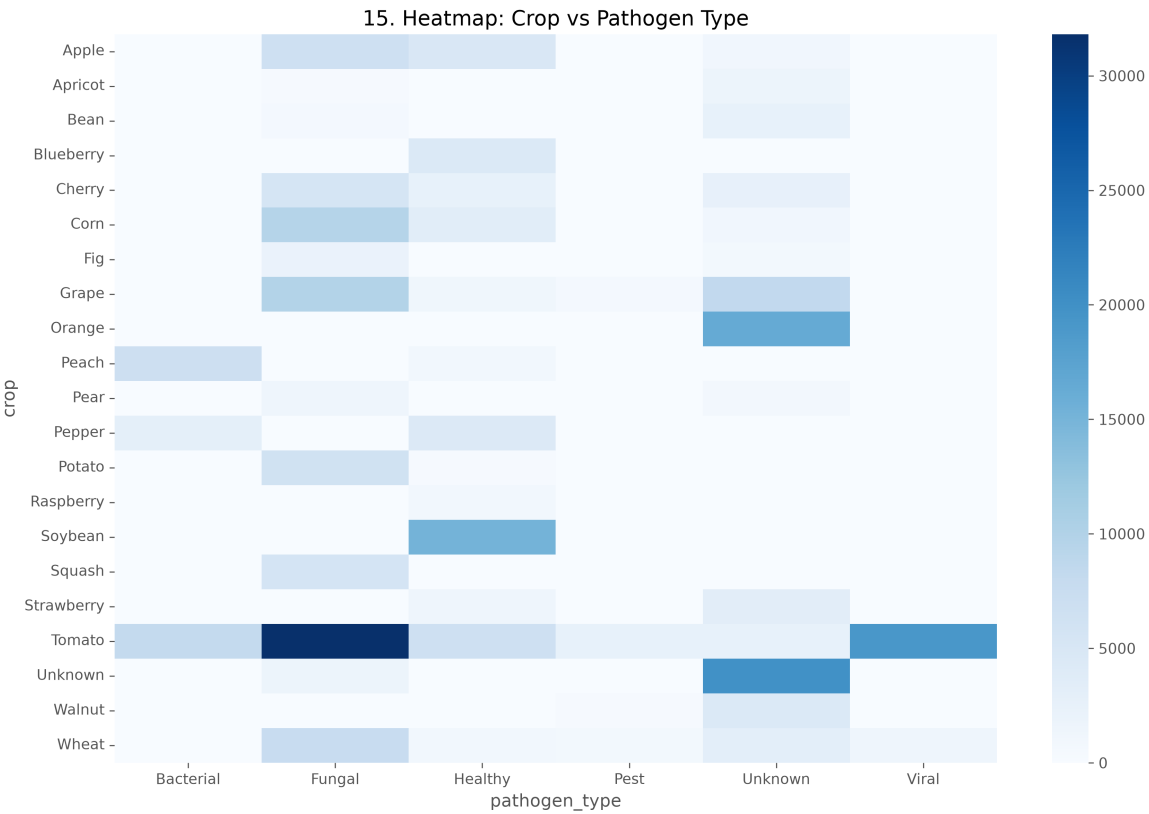


Figure: Crop Species x Pathogen Type Cross-Tabulation Matrix

18. Pathogen Types

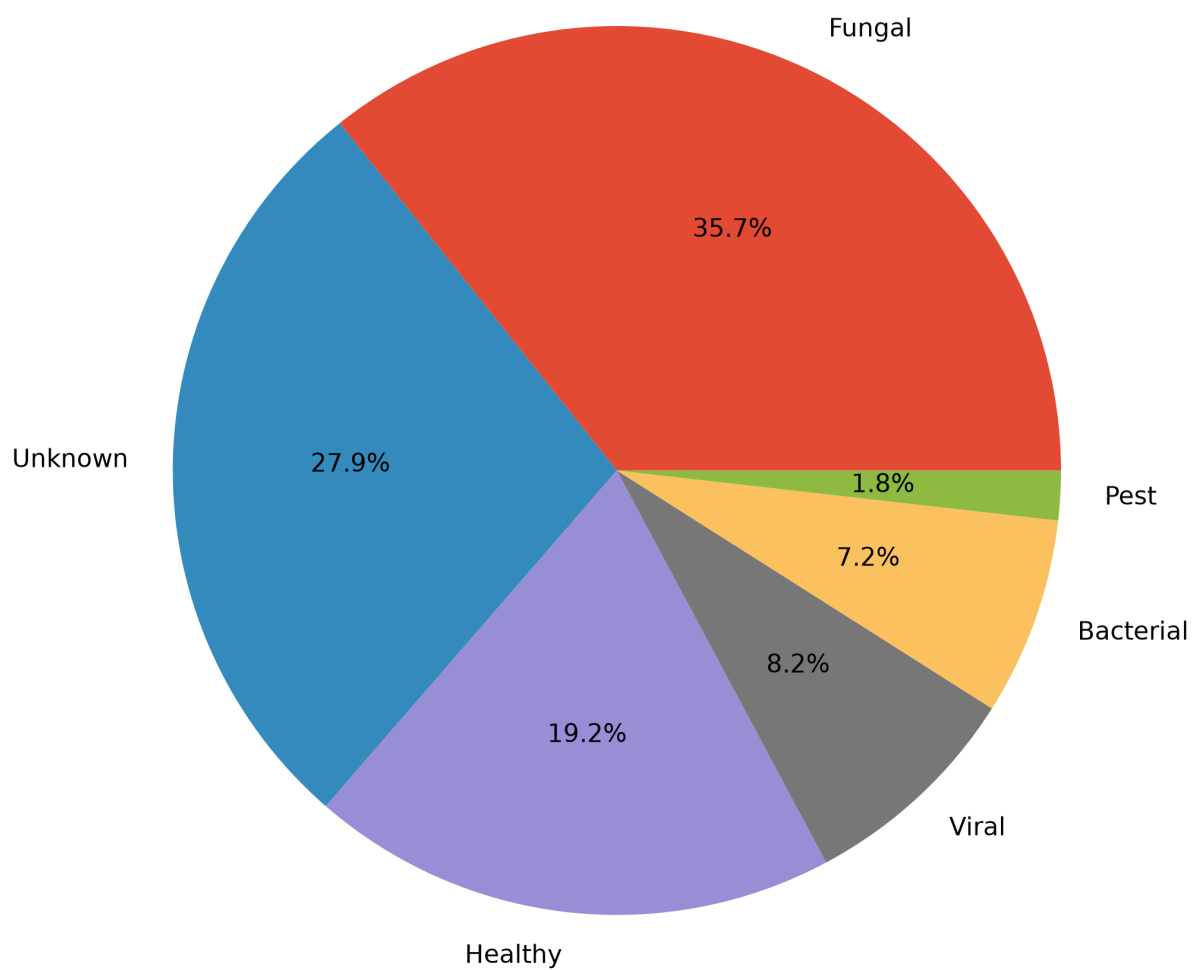


Figure: Pathogen Taxonomy Breakdown (Fungal, Bacterial, Viral, Oomycete)

3. Production 3-Crop Target Dataset EDA (Tomato, Potato, Pepper)

To maximize diagnostic precision for Pakistan's primary nightshade crops, ZARI.ai refined the dataset scope to Tomato (13 classes), Potato (3 classes), and Pepper (6 classes):

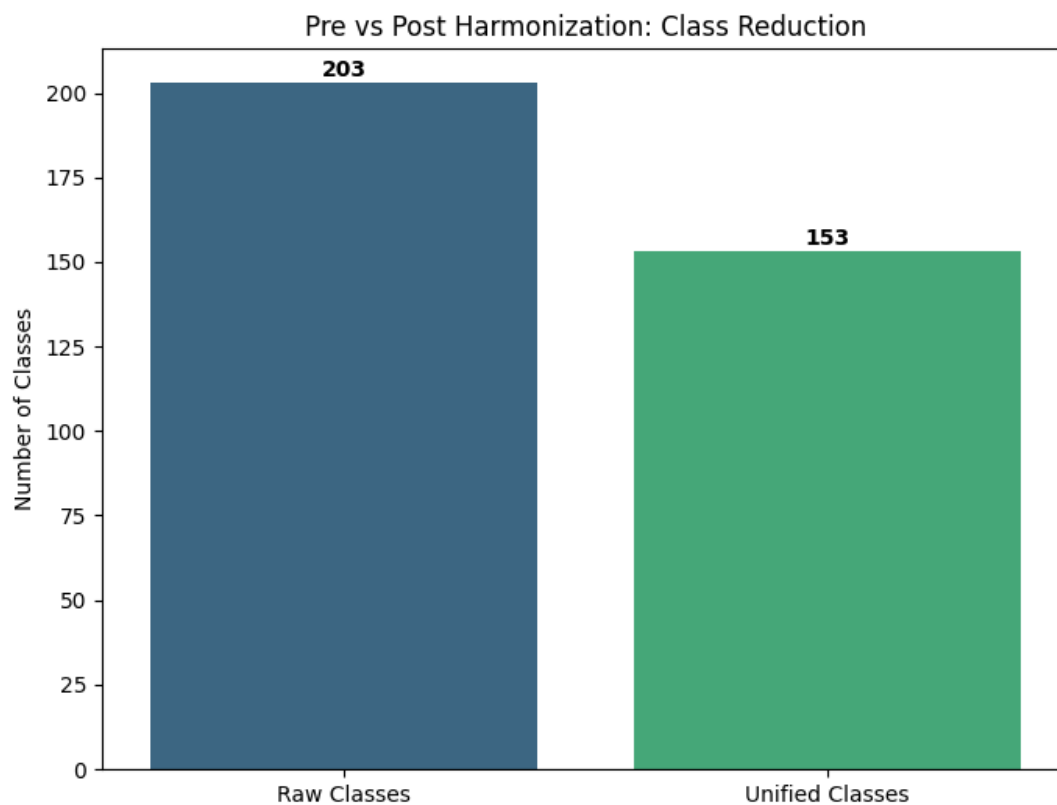


Figure: Class Scope Reduction: Filtering from 67 Classes down to 22 Target 3-Crop Classes

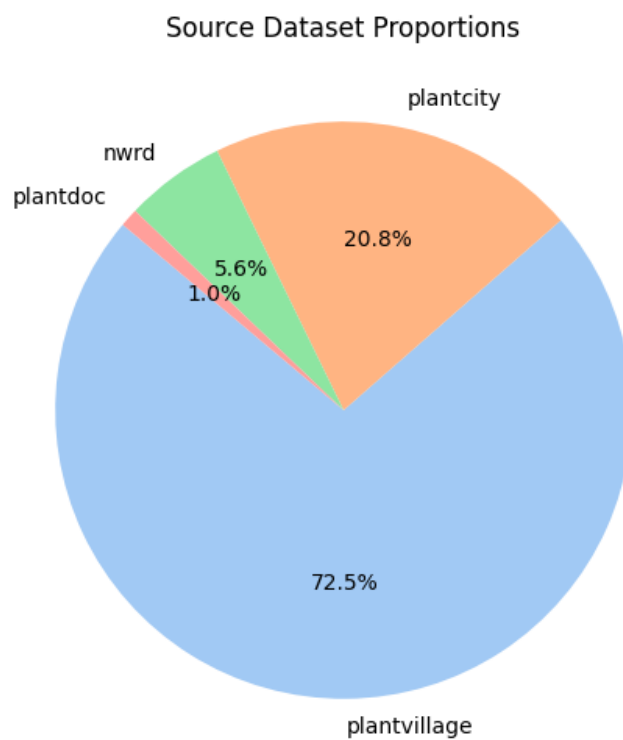


Figure: Data Provenance for 3 Target Crops

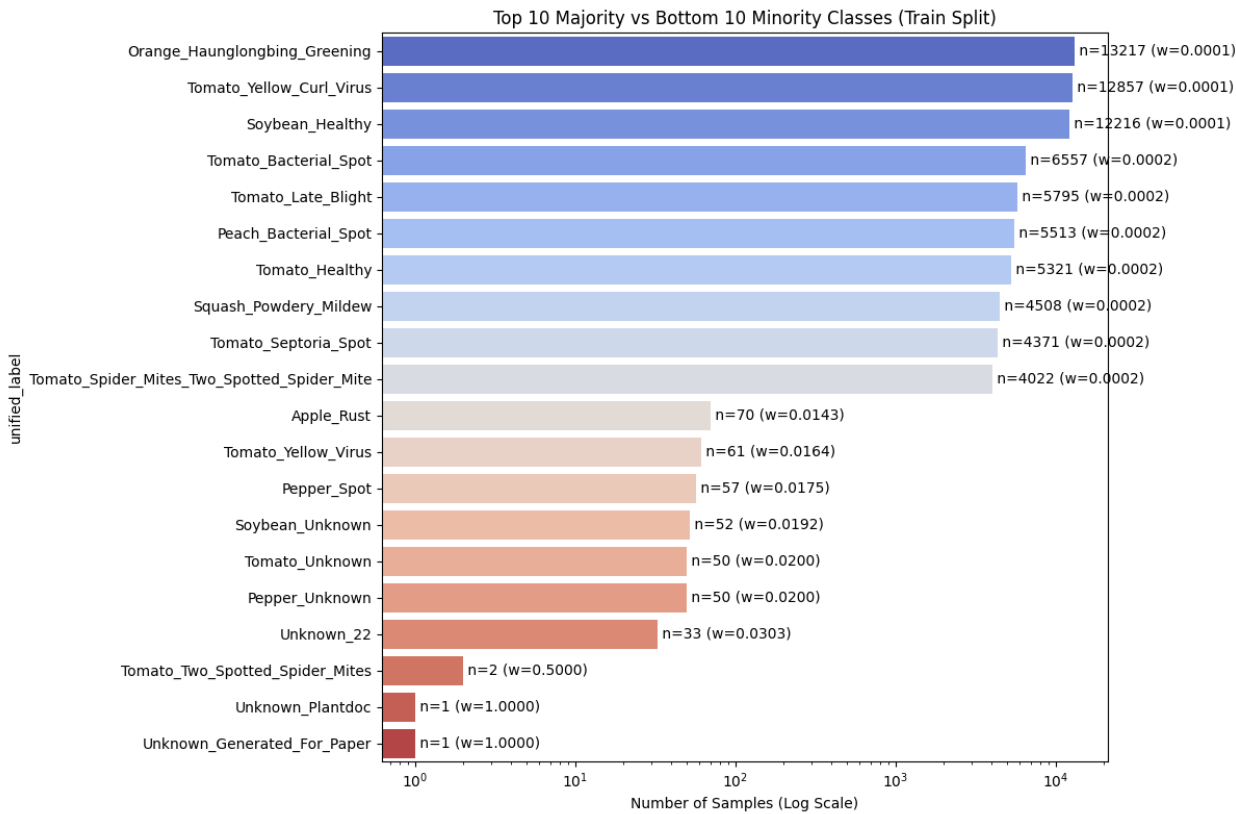


Figure: Class Imbalance Ratios Across 3 Target Crops (Tomato, Potato, Pepper)

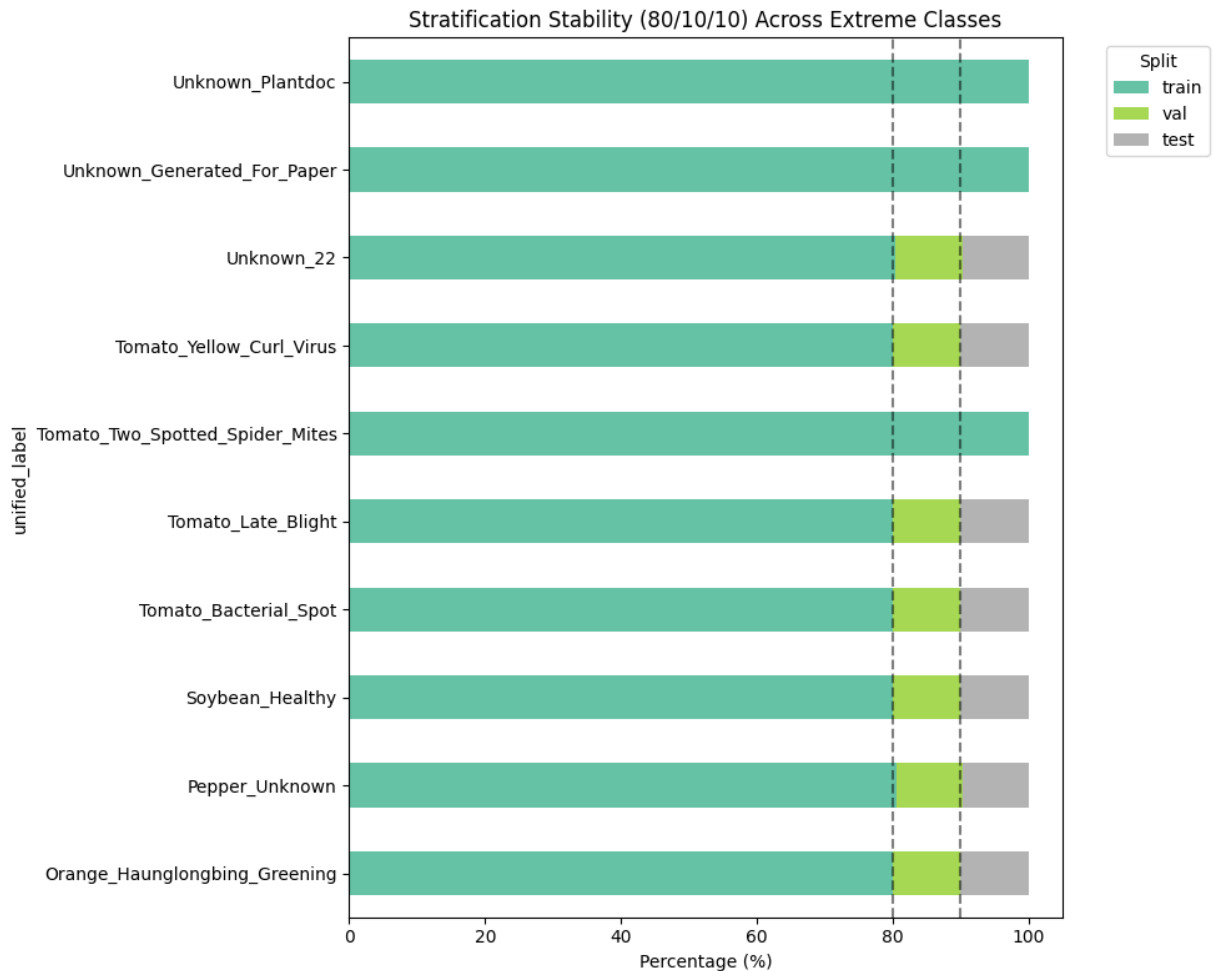


Figure: GroupKFold Stratified Split Verification without Data Leakage